Counterexamples to Lorentz's parity conjectures with graph dimension one

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Abstract

We construct a continuous nonodd function, normalized to vanish at zero, whose best uniform polynomial approximants agree in every pair of degrees $(2k+1, 2k+2)$, and a continuous noneven function whose best approximants agree in every pair $(2k, 2k+1)$. Both graphs have Hausdorff and box dimension exactly one. The baseline series has coefficients $(j+t)(-1/3)^j$. The factor $j+t$ creates a logarithmic gap at its extrema while preserving dimension one. A family of functional equations, with parameter shift $t \mapsto t+1$, propagates finite minimax certificates to every degree. The finite parts are specified by algebraic linear systems of orders 43 and 29 and verified by independent programs using exact rational arithmetic.

1 The assertions and the construction

Let $I = [-1, 1]$, and let $P_n f$ denote the unique best approximation to $f \in C(I)$ from the real polynomials of degree at most n, in the uniform norm. Write

$$T_k(\cos\theta) = \cos(k\theta), \qquad \Gamma_f = \{(x, f(x)) : x \in I\}.$$

The normalized oddness and evenness implications considered here are Conjectures 3′ and 4 in Saff and Varga [1].

Theorem 1.1. *There exist real continuous functions $f_{\mathrm{o}}, f_{\mathrm{e}}$ on I such that*

$$f_{\mathrm{o}}(0) = 0, \qquad P_{2k+1} f_{\mathrm{o}} = P_{2k+2} f_{\mathrm{o}} \quad (k \geq 0), \qquad f_{\mathrm{o}} \text{ is not odd},$$
$$P_{2k} f_{\mathrm{e}} = P_{2k+1} f_{\mathrm{e}} \quad (k \geq 0), \qquad f_{\mathrm{e}} \text{ is not even},$$

and

$$\dim_H \Gamma_{f_{\mathrm{o}}} = \dim_B \Gamma_{f_{\mathrm{o}}} = \dim_H \Gamma_{f_{\mathrm{e}}} = \dim_B \Gamma_{f_{\mathrm{e}}} = 1.$$

They can be taken to be

$$f_{\mathrm{o}}(x) = \sum_{j \geq 0} (j+1)(-1/3)^j T_{3^{j+1}}(x) + 10^{-10} \sum_{j \geq 0} (-1/3)^j h_{\mathrm{o}}(T_{3^j}(x)), \tag{1}$$

$$f_{\mathrm{e}}(x) = \sum_{j \geq 0} (j+1)(-1/3)^j T_{6 \cdot 3^j}(x) + 10^{-10} \sum_{j \geq 0} (-1/3)^j h_{\mathrm{e}}(T_{3^j}(x)), \tag{2}$$

where the degree-eleven polynomial h_{o} and the degree-ten polynomial h_{e} are defined exactly by the systems in Sections 5.1 and 5.2.

The role of the factor $j+1$ is analytic. With purely geometric critical coefficients, the baseline has only a linear gap at a contact, whereas an unrestricted perturbation can vary by $\delta \log(1/\delta)$. The extra factor strengthens the gap to this latter order. It changes the global modulus only to $O(\delta \log^2(1/\delta))$, which still gives graph dimension one. The proof below separates this analytic mechanism from the finite algebraic construction.

2 Critical series and their contact estimates

The purpose of this section is to retain dimension one while creating enough room below the extremal values to accommodate an arbitrary polynomial perturbation. Put

$$b = \frac{1}{3}, \qquad u = \frac{1}{\sqrt{2}}, \qquad G = \sum_{j=0}^{\infty} (-b)^j T_{3^j}, \qquad L = \sum_{j=0}^{\infty} j(-b)^j T_{3^j},$$

and, for every real $t \geq 1$, define

$$A_t = L + tG = \sum_{j=0}^{\infty} (j+t)(-b)^j T_{3^j}. \tag{3}$$

For a real polynomial h, an integer $d \geq 1$, and $\varepsilon > 0$, set

$$H = \mathcal{S}h := \sum_{j=0}^{\infty} (-b)^j h \circ T_{3^j}, \qquad F_t = A_t \circ T_d, \qquad f_t = F_t + \varepsilon H. \tag{4}$$

These series converge uniformly on $[-1, 1]$, since the Chebyshev polynomials map this interval onto itself and $\sum_{j\geq 0}(j+t)b^j < \infty$. Thus all these functions are continuous. Chebyshev composition and an index shift give

$$A_t = tT_1 - bA_{t+1} \circ T_3, \qquad H = h - bH \circ T_3, \qquad f_t = tT_d + \varepsilon h - bf_{t+1} \circ T_3. \tag{5}$$

The shift from t to $t+1$ will be handled by proving every initial approximation assertion uniformly for $t \geq 1$.

Lemma 2.1. *The function G is odd, has its unique maximum at u and its unique minimum at $-u$, and satisfies*

$$B := \|G\|_\infty = \frac{3u}{2} > 1, \qquad B - sG(y) \geq c_G|y - su| \quad (s = \pm 1), \qquad c_G = \frac{\pi(\sqrt{2}-1)}{324} > \frac{1}{300}. \tag{6}$$

Proof. Pair consecutive terms and put

$$g(x) = x - \frac{1}{3}T_3(x) = 2x - \frac{4}{3}x^3, \qquad G = \sum_{k\geq 0} 9^{-k} g \circ T_{9^k}.$$

The identity

$$g(u) - g(y) = \frac{4}{3}(y-u)^2(y+2u) \geq C_0(y-u)^2, \qquad C_0 = \frac{4}{3}(\sqrt{2}-1),$$

shows that g has its unique maximum at u. Since $T_9(u) = u$, every summand is maximized there, and the first summand makes the maximum unique. Summing their values gives $G(u) = 3u/2$. Oddness follows from the odd Chebyshev degrees.

For the positive-sign gap, write $y = \cos(\pi/4 + v)$ and put $\delta_0 = \pi/12$. If $0 < |v| \leq \delta_0$, choose the largest integer $k \geq 0$ with $9^k|v| \leq \delta_0$. Because $9^k \equiv 1 \pmod 8$, and sine is at least $1/2$ on $[\pi/6, \pi/3]$, the corresponding nonnegative deficit is at least

$$\frac{C_0}{4} 9^{-k}(9^k v)^2 > \frac{C_0\delta_0}{36}|v| \geq c_G|y-u|.$$

If $|v| > \delta_0$, then $|y-u| \geq r_0 = (\sqrt{3}-\sqrt{2})/2$, and the first summand gives the same conclusion: $r_0 > 1/8 > \pi/432 = c_G/C_0$. The case $v = 0$ is immediate. Reflection proves the negative-sign assertion. Finally, $\pi > 3$ and $\sqrt{2} > 7/5$ give $c_G > 1/270 > 1/300$. □

Use the increasing modulus

$$\rho(0) = 0, \qquad \rho(s) = s(1 + \log^+(1/s)) \quad (s > 0), \qquad \log^+ r = \max(\log r, 0). \tag{7}$$

In particular, $\rho(s) \geq s$.

Lemma 2.2. *For $t \geq 1$, the unique extrema of A_t are at $\pm u$, and*

$$B_t := \|A_t\|_\infty = B\left(t + \frac{1}{2}\right).$$

For $s = \pm 1$, put $\mathcal{A}_s = \{\theta \in \mathbb{R} : \cos\theta = su\}$. Then

$$B_t - sA_t(\cos\theta) \geq \frac{1}{4096}\rho(\operatorname{dist}(\theta, \mathcal{A}_s)) \qquad (\theta \in \mathbb{R},\ t \geq 1). \tag{8}$$

Proof. Let $U_j = (-1)^j G \circ T_{3^j}$. Absolute convergence permits summing the geometric convolution to obtain

$$L = \sum_{j \geq 1} b^j U_j.$$

Every U_j has norm B and takes the value B at u, because $T_3(u) = -u$. Thus $A_t = tG + \sum_{j\geq 1} b^j U_j$ has norm $B(t + 1/2)$. The positive coefficient of G makes the extrema unique.

First consider $\theta = \pi/4 + v$, where $0 < |v| \leq \delta_0 = \pi/12$. Let J be the largest integer with $3^J|v| \leq \delta_0$. Since 3^j is congruent to 1 or 3 modulo 8,

$$U_j(\cos(\pi/4 + v)) = G(\cos(\pi/4 + (-1)^j 3^j v)).$$

By Lemma 2.1, each term with $1 \leq j \leq J$ contributes at least $(c_G/2)|v|$ to the deficit of A_t. The single copy of G available because $t \geq 1$ supplies the same bound for index zero. Consequently,

$$B_t - A_t(\cos(\pi/4 + v)) \geq \frac{c_G}{2}(J + 1)|v|.$$

The inequalities $\delta_0 > 1/4$, $\log 4 < 2$, and $\log 3 < 2$ give $1 + \log(1/|v|) < 5(J + 1)$. The resulting bound is stronger than $\rho(|v|)/3000$.

For the remaining angles use

$$|\cos\theta - su| \geq \kappa \operatorname{dist}(\theta, \mathcal{A}_s), \qquad \kappa = \frac{4(1-u)}{\pi} > \frac{1}{4}. \tag{9}$$

For completeness, reduce the positive-sign case to $0 \leq \theta \leq \pi$. On $[0, \pi/4]$, the concavity chord for cosine proves the bound. On $[\pi/4, \pi/2]$, use $\sin\theta \geq u > \kappa$. On $[\pi/2, \pi]$, the convexity chord gives

$$u - \cos\theta \geq \frac{2\theta}{\pi} - (1 - u) \geq \kappa(\theta - \pi/4),$$

where the last inequality uses $u \geq 1/2$. Reflection proves the negative-sign case. If $r = \operatorname{dist}(\theta, \mathcal{A}_s) > \delta_0 > 1/4$, then $\rho(r) < 3r$. The contribution of G, together with (6) and (9), therefore gives a bound stronger than $\rho(r)/3600$. Periodicity and reflection reduce all remaining neighborhoods to the local case already proved. Both bounds imply (8). □

Lemma 2.3. *Put $\Phi = h \circ \cos$, $M = \|\Phi\|_\infty$, and $L_\Phi = \|\Phi'\|_\infty$. If v is a polynomial and $L_v = \|(v \circ \cos)'\|_\infty$, then*

$$|(H - v)(\cos\theta) - (H - v)(\cos\eta)| \leq (L_\Phi + 3M + L_v)\rho(|\theta - \eta|). \tag{10}$$

For Chebyshev expansions $h = \sum h_k T_k$ and $v = \sum v_k T_k$, the constants may be bounded by

$$M \leq \sum |h_k|, \qquad L_\Phi \leq \sum k|h_k|, \qquad L_v \leq \sum k|v_k|. \tag{11}$$

Proof. For $0 < \delta = |\theta - \eta| \le 1$, choose $J = \lfloor \log_3(1/\delta) \rfloor$. The terms of H with indices zero through J contribute at most $L_\Phi \delta (J+1)$, and the remaining terms contribute at most $M 3^{-J} < 3M\delta$. Since $\log 3 > 1$, $J + 1 \le 1 + \log(1/\delta)$. For $\delta \ge 1$, use $2\|H\|_\infty \le 3M \le 3M\delta$. The polynomial correction contributes at most $L_v \delta$, proving the assertion. The coefficient bounds follow from $T_k(\cos\theta) = \cos(k\theta)$. □

3 From finite contacts to all approximation degrees

We now identify the baseline approximants and show that a finite algebraic certificate works uniformly for the entire family $t \ge 1$.

Lemma 3.1 (Chebyshev composition)**.** *For $V \in C[-1,1]$, integers $m \ge 1$ and $n \ge 0$,*

$$P_n(V \circ T_m) = (P_{\lfloor n/m \rfloor} V) \circ T_m.$$

Consequently, if $D \ge \max(d, \deg h)$, then

$$P_n f_t = tT_d + \varepsilon h - b(P_{\lfloor n/3 \rfloor} f_{t+1}) \circ T_3, \qquad n \ge D. \tag{12}$$

Proof. Write a polynomial competitor as $p(\cos\theta) = \sum_{k=0}^{n} c_k \cos(k\theta)$, and average over the shifts $\theta + 2\pi j/m$, $0 \le j < m$. The target $V(\cos m\theta)$ is unchanged. Averaging cannot increase the uniform error, and removes every frequency not divisible by m. The average is therefore $r(\cos m\theta)$, with $\deg r \le \lfloor n/m \rfloor$. Surjectivity of T_m identifies the errors, while the reverse bound follows by composing $P_{\lfloor n/m \rfloor} V$ with T_m. Uniqueness of the best approximant gives the identity. Polynomial translation and nonzero scalar multiplication preserve the minimization problem by a bijection of competitors. Applying these operations to (5) proves (12). □

Lemma 3.2. *For $n < d$, set $r(n) = 0$. Otherwise let $r = r(n) \ge 1$ be the unique integer satisfying $d3^{r-1} \le n < d3^r$. Then*

$$p_n^0(t) := P_n F_t = \sum_{j=0}^{r-1} (j+t)(-b)^j T_{d3^j}, \tag{13}$$

$$F_t - p_n^0(t) = (-b)^r A_{t+r} \circ T_{d3^r}, \tag{14}$$

$$E_r(t) := \|F_t - p_n^0(t)\|_\infty = 3^{-r} B\left(t + r + \frac{1}{2}\right). \tag{15}$$

An empty sum is zero. Put $m = d3^r$. The full contact set is the set of zeros of T_{2m}, namely

$$x_j = -\cos\frac{(2j+1)\pi}{4m}, \qquad \sigma_j = (-1)^{r+m} \operatorname{sgn}\cos\frac{(2j+1)\pi}{4}, \qquad 0 \le j < 2m, \tag{16}$$

where σ_j is the residual sign. Positions and signs are independent of t. If $\Theta_{r,s} \subset [0,\pi]$ is the set of angular contacts of sign s, then

$$E_r(t) - s(F_t - p_n^0(t))(\cos\theta) \ge \frac{3^{-r}}{4096} \rho(\operatorname{dist}(\theta, \Theta_{r,s})). \tag{17}$$

Proof. Oddness gives $P_0 A_t = 0$. Repeatedly apply the recurrence obtained from $A_t = tT_1 - bA_{t+1} \circ T_3$ until the reduced degree is zero, then compose with T_d using Lemma 3.1. This proves (13); shifting the series proves (14). Lemma 2.2 gives (15) and the contact equation $T_m(x) = \pm u$, equivalent to $T_{2m}(x) = 0$. Evaluating T_m at the displayed nodes gives their signs.

For (17), apply (8) to the multiplied angle $m\theta$ and the sign $(-1)^r s$. The angular distance there is $m\,\mathrm{dist}(\theta, \Theta_{r,s})$. Indeed, the endpoints of $[0, m\pi]$ are integer multiples of π; reflection replaces any exterior nearest periodic target by an equally close interior target. Multiplication by 3^{-r}, and monotonicity of ρ, prove the stated bound. □

Proposition 3.3 (Persistence of finite certificates)**.** *Let $\mathcal{N}$ be a nonempty finite set of positive upper degrees. Assume $\deg p_N^0(t) \le N-1$ for $N \in \mathcal{N}$. For each N, suppose a polynomial v_N of degree at most $N-1$ and a real number λ_N satisfy, at every contact in (16) for level $r(N)$,*

$$\sigma_j(H(x_j) - v_N(x_j)) \le \lambda_N, \tag{18}$$

with equality at $N+2$ ordered contacts whose signs alternate. Put

$$R = \max_{N\in\mathcal{N}} r(N), \qquad C = L_\Phi + 3M + \max_{N\in\mathcal{N}} L_{v_N}, \qquad \Lambda = \max_{N\in\mathcal{N}} |\lambda_N|.$$

If

$$0 < \varepsilon \le \min\left\{ \frac{3^{-R}}{8192(1+C)}, \frac{3^{-R}}{2(1+\Lambda)} \right\}, \tag{19}$$

then, simultaneously for every $t \ge 1$ and $N \in \mathcal{N}$,

$$P_N f_t = P_{N-1} f_t = p_N^0(t) + \varepsilon v_N.$$

Proof. Fix N, put $r = r(N)$, and fix a sign s. For any angle θ, choose a nearest contact $\eta \in \Theta_{r,s}$, and put $\delta = |\theta - \eta|$. Equations (18), (10), and (17) give

$$\begin{aligned} s(f_t - p_N^0(t) - \varepsilon v_N)(\cos\theta) &\le E_r(t) + \varepsilon\lambda_N - \left(\frac{3^{-r}}{4096} - \varepsilon C\right)\rho(\delta) \\ &\le E_r(t) + \varepsilon\lambda_N. \end{aligned}$$

Apply both signs to obtain the norm bound. Since $B > 1$ and $t \ge 1$, (15) gives $E_r(t) > 3^{-R}$; thus (19) also makes the corrected error level positive. At the selected $N+2$ contacts there is exact alternating equality. The alternation theorem identifies $P_N f_t$. This polynomial has degree at most $N-1$, so it is also optimal in that smaller space; uniqueness identifies $P_{N-1} f_t$. □

Proposition 3.4 (Induction over all parameters)**.** *Suppose the hypotheses of Proposition 3.3 hold.*

1. *If d is odd, $D = 2m+1 \ge \max(d, \deg h)$ with $m \ge 1$, and $\mathcal{N} = \{2, 4, \dots, 2m\}$, then $P_{2k+1} f_t = P_{2k+2} f_t$ for all $k \ge 0$ and $t \ge 1$.*
2. *If d is even, $D = 2m \ge \max(d, \deg h)$ with $m \ge 1$, and $\mathcal{N} = \{1, 3, \dots, 2m-1\}$, then $P_{2k} f_t = P_{2k+1} f_t$ for all $k \ge 0$ and $t \ge 1$.*

In these two cases the baseline degree assumption in Proposition 3.3 holds automatically.

Proof. Every frequency $d3^j$ in (13) has the parity of d, so the baseline prefix at each upper degree already has degree at most one less. For the first assertion use strong induction on the odd lower degree n, with the induction statement quantified over every $t \ge 1$. The initial cases follow from Proposition 3.3. For odd $n \ge D$, apply (12) to n and $n+1$. Their floor indices agree unless $n = 6k+5$; in that case they are $2k+1, 2k+2$, a smaller pair covered by induction at parameter $t+1$. The second assertion is identical with even lower degrees: the exceptional case is $n = 6k+2$, giving the smaller pair $2k, 2k+1$. □

4 Dimension of the constructed graphs

Proposition 4.1. *For every polynomial h, integer $d \geq 1$, real $t \geq 1$, and $\varepsilon > 0$, the function f_t in (4) satisfies*

$$\dim_H \operatorname{graph} f_t = \dim_B \operatorname{graph} f_t = 1.$$

Here $\dim_B$ denotes the box dimension; its lower and upper versions coincide for these graphs.

Proof. In the angular coordinate the baseline is

$$F_t(\cos\theta) = \sum_{j\geq 0} (j+t)(-b)^j \cos(d3^j\theta).$$

For $0 < \delta \leq 1/2$, split this series at $J = \lfloor \log_3(1/\delta) \rfloor$. The oscillation of its head over an angular displacement at most δ is bounded by $d\delta \sum_{j\leq J}(j+t)$. The tail is bounded by $2\sum_{j>J}(j+t)3^{-j} = O_t(\delta(J+t+1))$. Thus, for fixed t, d, the baseline modulus is $O_{t,d}(\delta(1+\log(1/\delta))^2)$. Lemma 2.3 bounds the perturbation by a smaller modulus. Consequently, the graph of $\theta \mapsto f_t(\cos\theta)$ over $[0,\pi]$ can be covered by

$$O\Big(\delta^{-1}(1+\log(1/\delta))^2\Big)$$

squares of side δ: use intervals of length δ in the horizontal coordinate and the preceding oscillation bound in each interval. Its upper box dimension is at most one. The Lipschitz map $(\theta, y) \mapsto (\cos\theta, y)$ maps this graph onto graph f_t, whose upper box dimension is therefore at most one as well. Projection onto the horizontal interval $[-1,1]$ gives Hausdorff dimension at least one, proving both equalities. □

5 How the algebraic contact systems are specified

Set $\omega = -1/3$. For a positive integer M define the ordered grid

$$x_j^{(M)} = -\cos\frac{(2j+1)\pi}{2M}, \qquad 0 \leq j < M.$$

For a baseline composition degree d and a level r, take $M = 2d3^r$. The residual sign at this node is

$$\sigma_{r,M}(j) = (-1)^{r+M/2} \operatorname{sgn}\cos\frac{(2j+1)\pi}{4}. \tag{20}$$

Given a forcing h and its critical series $H = \mathcal{S}h$, a correction polynomial v, and a scalar λ, a contact equation means

$$\sigma_{r,M}(j)\big(H(x_j^{(M)}) - v(x_j^{(M)})\big) = \lambda. \tag{21}$$

The slack at any node is the right side minus the left side.

All these equations are finite algebraic equations. To make this explicit, write $h = \sum c_k T_k$ and evaluate the series for one basis term. For the odd grids, put $J = r+1$; after J iterations the node reaches the two-cycle $\{1/\sqrt{2}, -1/\sqrt{2}\}$. Thus

$$(\mathcal{S}T_k)(x) = \sum_{i=0}^{J-1} \omega^i T_{k3^i}(x) + \frac{\omega^J T_{k3^J}(x)}{1-\omega(-1)^k}. \tag{22}$$

For the even grids, put $J = 1$ when $M = 12$ and $J = 2$ when $M = 36$. The eventual angular orbit has period dividing four, giving

$$(\mathcal{S}T_k)(x) = \sum_{i=0}^{J-1} \omega^i T_{k3^i}(x) + \frac{\sum_{i=J}^{J+3} \omega^i T_{k3^i}(x)}{1-\omega^4}. \tag{23}$$

These are exact sums of the infinite series, not truncations.

5.1 The normalized oddness example

Write

$$h_{\mathrm{o}} = c_2(T_2+1) + c_4(T_4-1) + c_6(T_6+1) + c_8(T_8-1) + c_{10}(T_{10}+1) + c_5T_5 + c_7T_7 + c_{11}T_{11}. \tag{24}$$

In particular $h_{\mathrm{o}}(0) = 0$. For $N \in \{2,4,6,8,10\}$ introduce a correction v_{N-1} of degree at most $N-1$ and a scalar λ_N. The eight forcing coordinates, thirty correction coefficients, and five error corrections give 43 unknowns. Impose (21) at every index in the last column of Table 1, and impose

$$c_6 = 1, \qquad c_5 = -40.$$

There are 41 contact equations and two normalizations. This square system uniquely defines all the unknowns.

Upper degree N	Level r	Grid size M	All equality indices
2	0	6	$0,2,3,5$
4	1	18	$0,2,7,10,15,17$
6	1	18	$0,1,3,5,12,14,15,17$
8	1	18	$0,1,3,5,7,10,12,14,16,17$
10	2	54	$1,3,7,10,15,18,23,30,35,38,43,46,52$

Table 1: Exact contact system for h_{o}. For $N = 10$, omit index 7 to obtain the twelve alternating contacts. Every other row already consists of $N+2$ alternating contacts.

Exact elimination and rational interval evaluation give:

1. every elimination pivot is nonzero;
2. all the displayed contacts have zero slack;
3. every other slack exceeds $1/3000$;
4. the indicated subsets have alternating residual signs;
5. the coefficient bounds in Table 3 hold.

The exact data are in `certificates/odd/critical_odd_certificate.json`. The independent checker reconstructs the actual periodic orbit sums and verifies these assertions without using the generator's arithmetic code.

For orientation only, the coefficients of h_{o} in the Chebyshev basis are

k	$[T_k]h_{\mathrm{o}}$	k	$[T_k]h_{\mathrm{o}}$
0	2.160663098060	1	0
2	−0.062874821143	3	0
4	−1.047297917263	5	−40
6	1	7	5.619841448418
8	−0.239114823082	9	0
10	−0.062874821143	11	−35.248063651996

These rounded numbers do not define the construction; the exact contact system does.

5.2 The evenness example

Write

$$h_{\mathrm{e}} = \sum_{k\in\{1,2,3,4,5,7,8,9,10\}} c_k T_k. \tag{25}$$

Introduce a constant w_0, a polynomial w_6 of degree at most six, a polynomial w_8 of degree at most eight, and three scalars μ_5, μ_7, μ_9. Together with the nine forcing coefficients, these give 29 unknowns. Impose (21) at all indices of Table 2, using the correction indicated there, and add $c_7 = -1$. There are 28 contact equations and one normalization; the resulting square system has a unique solution.

N	r	M	Correction	All equality indices
5	0	12	w_0	$0, 1, 4, 5, 7, 10, 11$
7	1	36	w_6	$1, 3, 4, 6, 11, 22, 24, 29, 31, 34$
9	1	36	w_8	$1, 3, 6, 11, 13, 16, 22, 24, 29, 31, 34$

Table 2: Exact contact system for h_{e}. For $N = 7$, omit index 4 to obtain nine alternating contacts. The other rows already alternate.

The exact checks prove that every inactive slack exceeds $1/4$, in addition to all pivot, contact, alternation, and coefficient assertions. They also prove the endpoint witness

$$-\frac{1}{20} < \sum_{\substack{1\le k\le 9\\ k \text{ odd}}} c_k < -\frac{1}{50}. \tag{26}$$

The exact data are in `certificates/even/critical_even_certificate.json`. Descriptive decimal coefficients are

k	$[T_k]h_{\mathrm{e}}$	k	$[T_k]h_{\mathrm{e}}$
0	0	1	−0.749781201737
2	37.546062458564	3	0.731626781646
4	−91.154700518253	5	0.731626781646
6	0	7	−1
8	52.992212214940	9	0.250218798263
10	141.341851838786		

5.3 Fields, root identification, and reproducibility

Define

$$M_{12}(z) = z^{12} - 12z^{10} + 54z^8 - 112z^6 + 105z^4 - 36z^2 + 1.$$

The odd system is computed in $\mathbb{Q}(z)$ with $z = 2\cos(\pi/108)$ and polynomial $M_{12}(z^3 - 3z)$. The even system uses $z = 2\cos(\pi/72)$ and polynomial $M_{12}(z^2 - 2)$. The relevant roots are isolated, respectively, in $(1999/1000, 2)$ and $(999/500, 2)$ by exact rational bisection. Their uniqueness follows from

$$M_{12} = S_6^2 - 3, \quad S_6(w) = w^6 - 6w^4 + 9w^2 - 2, \quad S_6'(w) = 6w(w^2 - 1)(w^2 - 3) > 0$$

on the relevant positive interval, and the monotonicity of $z^3 - 3z$ or $z^2 - 2$. The elementary bound $2\cos\theta \ge 2 - \theta^2$ identifies the designated trigonometric root. Chebyshev recurrence then constructs all node values algebraically.

The generators use FLINT for rational polynomial arithmetic. The two independent auditors use only the Python standard library, with separate implementations. The odd auditor uses integer numerator polynomials with a common denominator; the even auditor uses rational coefficient arrays. Both detect the actual eventual T_3 orbits, sum those orbits exactly, check every stored slack, and prove strict signs by rational intervals. No floating-point tolerance is an acceptance condition. The decimal tables and diagnostic output have no role in the proof.

Certified quantity	Odd example	Even example
$\sum_k \lvert[T_k]h\rvert$	< 100	< 400
$\sum_k k\lvert[T_k]h\rvert$	< 700	< 3000
$\max_v \sum_k \lvert[T_k]v\rvert$	< 200	< 400
$\max_v \sum_k k\lvert[T_k]v\rvert$	< 1500	< 3000
Absolute error correction	< 200	< 200
Every positive slack	$> 1/3000$	$> 1/4$
Largest initial level R	2	1

Table 3: Exact bounds used by the analytic persistence estimate.

6 Completion of the two constructions

Take $\varepsilon = 10^{-10}$ and use the notation of the analytic construction. For the odd certificate, the perturbation modulus constant is at most

$$C_{\mathrm{o}} < 700 + 3 \cdot 100 + 1500 = 2500.$$

For the even certificate,

$$C_{\mathrm{e}} < 3000 + 3 \cdot 400 + 3000 = 7200.$$

Consequently

$$\varepsilon C_{\mathrm{o}} < \frac{3^{-2}}{4096}, \qquad \varepsilon C_{\mathrm{e}} < \frac{3^{-1}}{4096}.$$

The corrected errors remain positive because the baseline errors exceed 3^{-R} and the absolute corrections are less than 200ε. The persistence argument therefore applies uniformly for every $t \geq 1$.

For $f_{\mathrm{o},t} = A_t \circ T_3 + \varepsilon \mathcal{S} h_{\mathrm{o}}$ it yields

degrees	common best approximant
$1, 2$	εv_1
$3, 4$	$tT_3 + \varepsilon v_3$
$5, 6$	$tT_3 + \varepsilon v_5$
$7, 8$	$tT_3 + \varepsilon v_7$
$9, 10$	$tT_3 - (t+1)T_9/3 + \varepsilon v_9$.

The forcing degree is at most eleven. Simultaneous induction in degree over all $t \geq 1$ gives every odd/even plateau. The form (24) gives $h_{\mathrm{o}}(0) = 0$, whence $(\mathcal{S}h_{\mathrm{o}})(0) = 0$ and $f_{\mathrm{o},t}(0) = 0$. The even part of h_{o} is nonzero because its T_6 coefficient is exactly one. The operator $\mathcal{S}$ preserves parity and has inverse $H \mapsto H + H \circ T_3/3$ on its range, so its even part remains nonzero. Since the baseline is odd, $f_{\mathrm{o},t}$ is not odd.

For $f_{\mathrm{e},t} = A_t \circ T_6 + \varepsilon \mathcal{S}h_{\mathrm{e}}$, the constant certificate at upper degree five gives

$$P_0 f_{\mathrm{e},t} = \cdots = P_5 f_{\mathrm{e},t} = \varepsilon w_0.$$

The other two certificates give

$$P_6 f_{\mathrm{e},t} = P_7 f_{\mathrm{e},t} = tT_6 + \varepsilon w_6, \qquad P_8 f_{\mathrm{e},t} = P_9 f_{\mathrm{e},t} = tT_6 + \varepsilon w_8.$$

The degree-ten forcing and simultaneous induction give all even/odd plateaus. Since T_3 fixes both endpoints, (26) gives

$$-\frac{3}{40}\varepsilon < f_{\mathrm{e},t}(1) - f_{\mathrm{e},t}(-1) < -\frac{3}{100}\varepsilon < 0.$$

Thus $f_{\mathrm{e},t}$ is not even. Taking $t = 1$ and applying the dimension result proved above establishes Theorem 1.1.

7 What is and is not asserted

The construction proves dimension one, both in the Hausdorff and in the box sense. It does not produce smooth or Lipschitz examples: the logarithmic contact gap is compatible with graph dimension one and is the reason the perturbation persists. The all-degree conclusion is analytic, not a finite numerical extrapolation.

The finite certificate verification is exact computer-assisted verification. This extension has not been formalized in Lean. To reproduce the finite checks from the source directory, run:

```
python3 certificates/odd/audit_critical_odd.py
python3 certificates/even/audit_critical_even.py
```

Regeneration additionally requires `python-flint`. The accompanying README gives the commands and distinguishes generation from verification.

References

[1] E. B. Saff and R. S. Varga, *Remarks on some conjectures of G. G. Lorentz*, Journal of Approximation Theory **30** (1980), 29–36. https://www.math.kent.edu/~varga/pub/paper_126.pdf.